Lab 7 Reflections

Khin Thu

Lab 7 helped me understand that evaluating a machine learning model involves more than looking at its accuracy. A single score may seem convincing, but it does not explain which predictions were incorrect or why those mistakes matter. Using the Titanic survival example helped me connect evaluation metrics to a clear problem. This is supervised learning because the model learns from passenger information with known survival outcomes. However, having labeled data does not guarantee that the model will make reliable predictions on new examples.

One important concept was the confusion matrix. Separating predictions into true positives, true negatives, false positives, and false negatives made model errors easier to understand. In this example, a false positive means predicting that a passenger survived when they did not, while a false negative means predicting that a passenger died when they actually survived. Understanding these categories helped me see why two models with similar accuracy could still behave differently. The types of errors they make may be more important than their overall number of correct predictions.

Precision, recall, and F1-score added another layer to my understanding. Precision focuses on how often positive predictions are correct, while recall focuses on how many actual positive cases are identified. Keeping these definitions separate requires attention because both describe performance from different perspectives. Connecting them to real situations makes the distinction clearer. For example, a cybersecurity system with low precision could produce many false alerts, creating extra work for analysts. A system with low recall could miss real attacks. F1-score balances precision and recall, but choosing an evaluation metric still requires understanding the consequences of each mistake.

Cross-validation also changed how I think about model reliability. A single train-test split can produce a result that depends heavily on which examples were selected. Five-fold cross-validation evaluates the model across several splits, providing a broader view of its performance. The average score is useful, but variation between folds also matters because it can reveal inconsistent results.

This lab connects to earlier work on preparing data and training models. I now understand evaluation as a step that guides the next decision in the workflow. Instead of accepting a model because one score looks good, I would examine its errors, compare several metrics, and consider whether its performance is consistent. My main takeaway is that a useful model must perform well in ways that match the needs of the problem.

Antonio Garcia

In this lab, I worked through the core ideas behind model evaluation, and it helped me understand how important it is to measure a model's performance correctly instead of just assuming it works. At first, the workflow felt straightforward because it was mostly about running evaluation metrics, but as I went through each step, I realized how much these metrics reveal about a model's strengths and weaknesses. Using train-test splits and cross-validation made me see that a single accuracy score doesn't tell the full story. A model might look good on one split but perform poorly on another, and cross-validation helps expose that inconsistency. This made me appreciate why machine learning workflows rely on multiple evaluation methods rather than a single number.

Working with metrics like accuracy, precision, recall, and F1-score also deepened my understanding of how different problems require different evaluation strategies. For example, accuracy alone can be misleading when classes are imbalanced, and seeing that in the outputs made the concept click more than just reading about it. The confusion matrix was especially helpful because it showed exactly where the model was making mistakes. Instead of thinking "the model is 90% accurate," I could see whether it was misclassifying certain categories more than others. That visual breakdown made the evaluation feel more concrete and practical.

I also learned how important it is to think about generalization. The lab made me compare training performance with testing performance, and whenever the gap was too large, it signaled overfitting. Even though this lab wasn't focused on bias-variance like Lab 08, the idea still showed up here because evaluation metrics reveal whether a model is learning patterns or just memorizing the training data. Seeing the numbers change depending on the dataset split helped me understand why evaluation is not just a final step but something that guides model selection and improvement.

Throughout the notebook, I ran into a few small issues, especially with making sure the outputs displayed correctly and that the metrics matched the expected format. Fixing those problems forced me to slow down and pay attention to how the data flowed through each cell. It reminded me that machine learning isn't just about writing code — it's about understanding what each step means and verifying that the results make sense. When something looked off, I had to question whether the model was behaving poorly or whether I made a mistake in the workflow. That habit of checking assumptions is something I'm starting to build through these labs.

Overall, this lab helped me see model evaluation as a critical part of machine learning rather than an optional add-on. It showed me that good evaluation prevents bad models from slipping through and gives a clearer picture of how well a model will perform on real-world data. I also realized how evaluation connects back to earlier labs, like how the

classification models from Lab 06 depend heavily on these metrics to judge their performance. Going forward, I'll be more intentional about choosing the right evaluation methods and making sure I understand what the numbers actually mean instead of just reporting them.

Steven Ford

Lab 7 completely changed how I think about measuring machine learning performance by moving beyond a single accuracy score to examine where and why models make mistakes. Working with the confusion matrix made the distinction between false positives and false negatives much more concrete, highlighting why metrics like precision and recall are critical depending on the specific costs of error in a given problem. Balancing these trade-offs with the F1-score showed me that a model's true effectiveness depends heavily on understanding the real-world consequences of its misclassifications rather than just chasing a high overall percentage.

Additionally, exploring cross-validation and generalization reinforced why relying on a single train-test split can be misleading due to dataset variance. Evaluating models across multiple folds provided a much clearer picture of stability and consistency, helping me realize that proper evaluation isn't just a final checkbox in a workflow, but a vital guide for model selection and iterative improvement.

Cristian Delgado

Module 7 goes over the confusion matrix as well as Recall and Precision. Recall as it relates to the confusion matrix measures how often it actually got the positive identification correct relative to the positive outcomes in total. Precision measure how often its result's positive prediction was correct. The module emphasis that both have their own tradeoffs, where recall benefits scenarios where false negatives can prove costly, and precision benefits those where false positives can prove costly, and these are usually balances through an F1 score. We also explored how cross validation helps improve our models by making sure its trains for the average general scenario, versus potentially focusing on dataset examples that lean to one side or another.

The lab setup was straightforward and provided a clear example of a confusion matrix as well as a few basic reports to review. One question I have is if its possible to have a high recall and high precision, or if focusing on 1 will have a negative consequence on the other.
